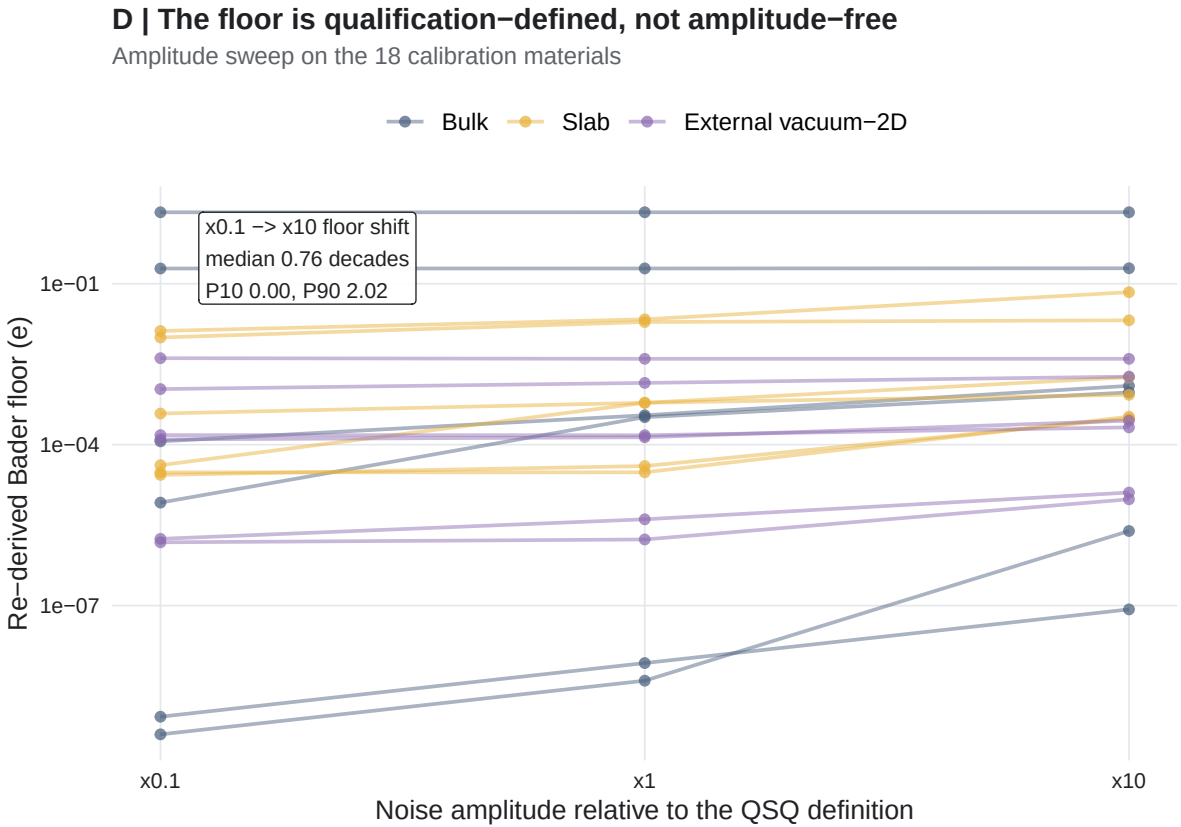
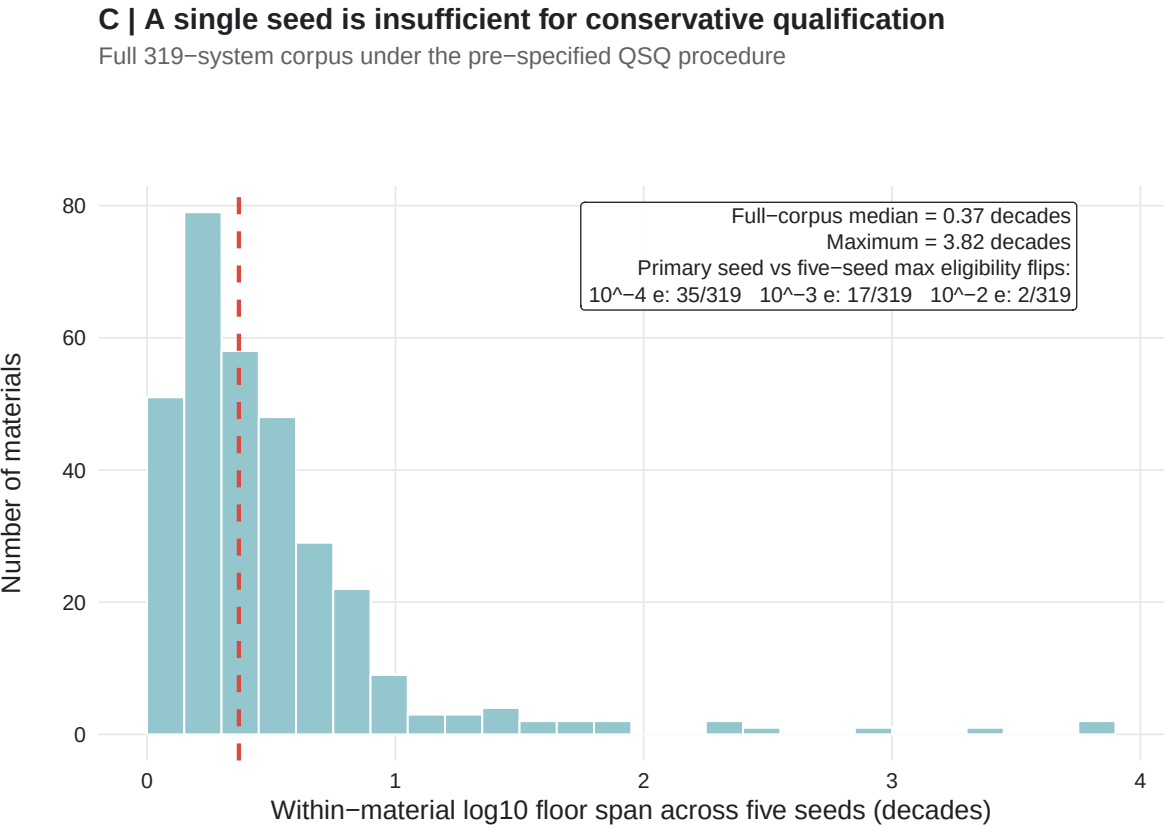
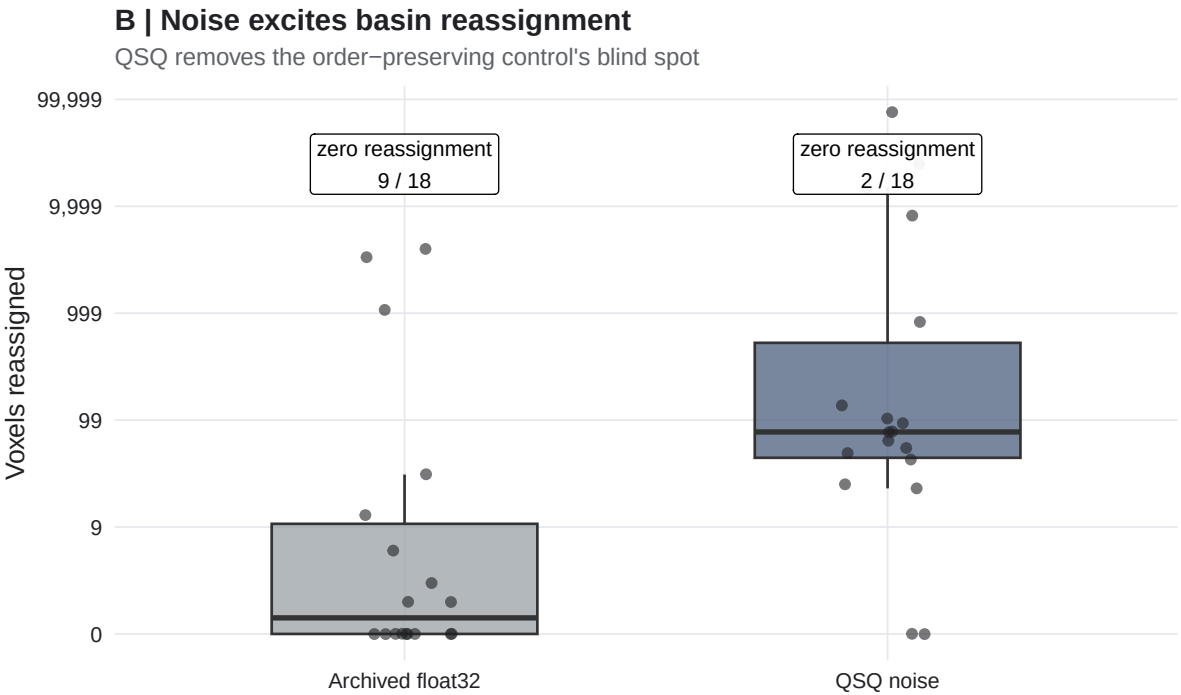
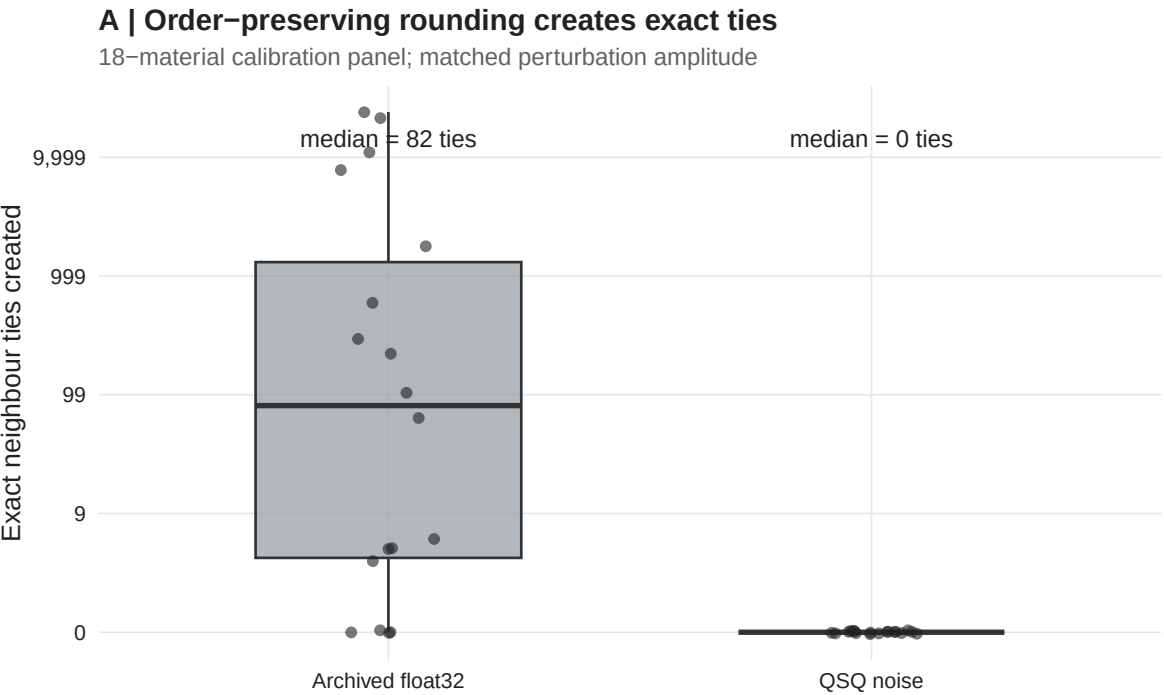


Supplementary Figure S2 | QSQ probe validation, seed sensitivity and amplitude sensitivity

The calibration justifies a non-order-preserving, multi-seed probe; full-corpus statistics quantify its behavior without retuning.



A–B, 18-material probe calibration. C, five-seed variability in all 319 systems. D, two-decade amplitude sensitivity. The QSQ amplitude and seed set were pre-specified before validation.